

Assignment1



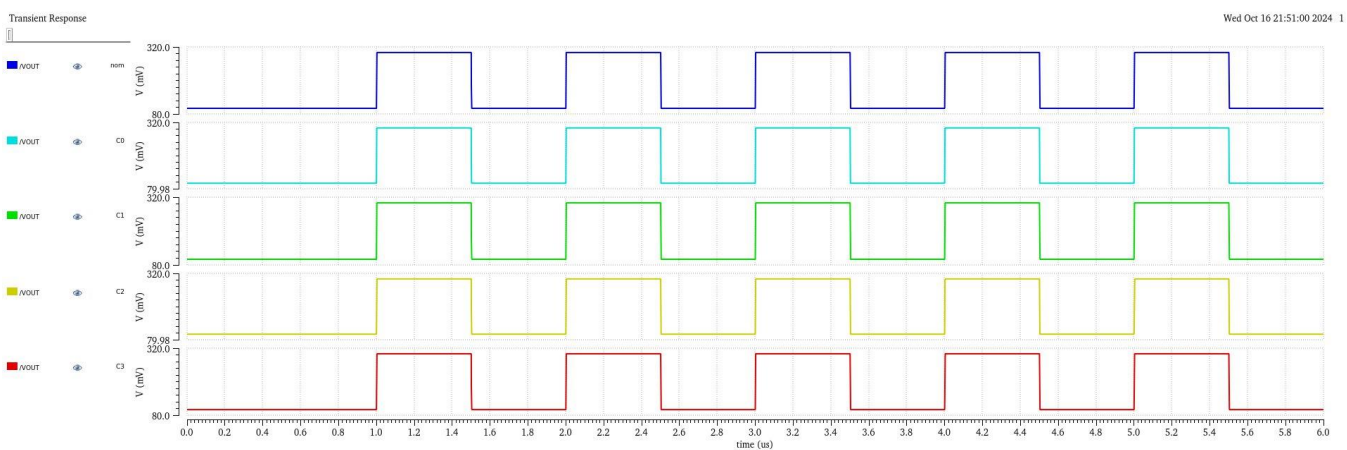
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Part I: MOSFET characteristics across Corners and Temperature

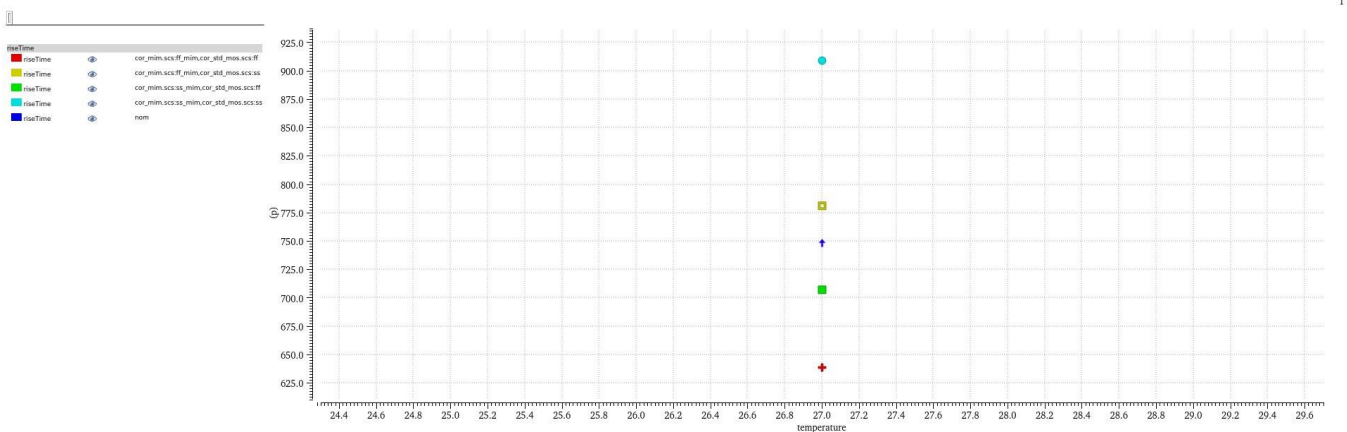
i. Transient Response

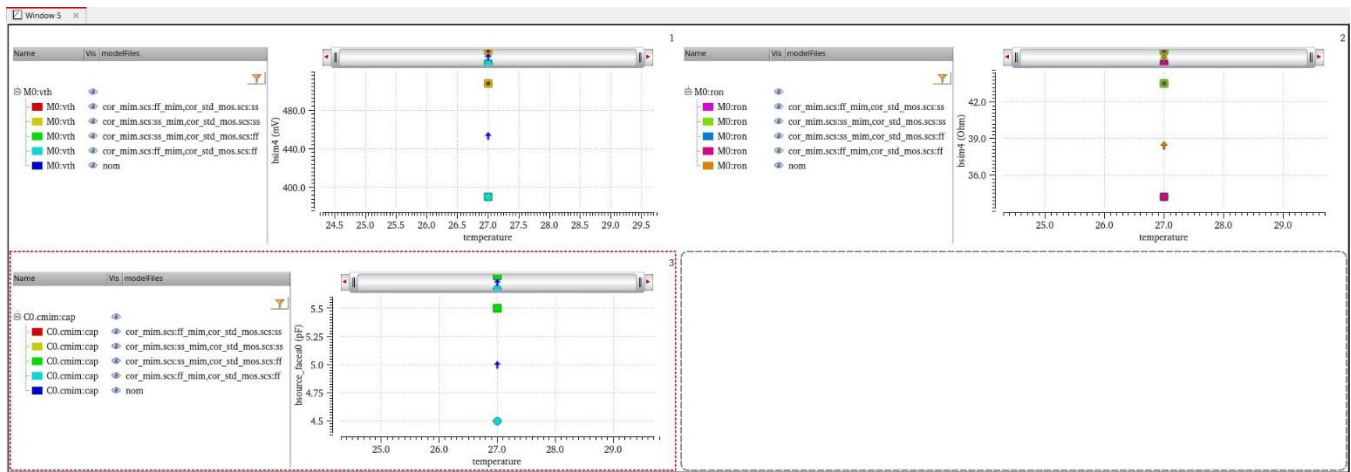
- **$V_G = V_{dd}$**
- **V_{in} is square wave (with average of 200mV) with period of 1 μ s and rise and fall time of 0.5ns**
- **Perform transient analysis with time of 5 μ s**
- **Plot V_{out} across corners**



- **Plot Rising time of V_{out} across corners and explain at each corner what happens?**

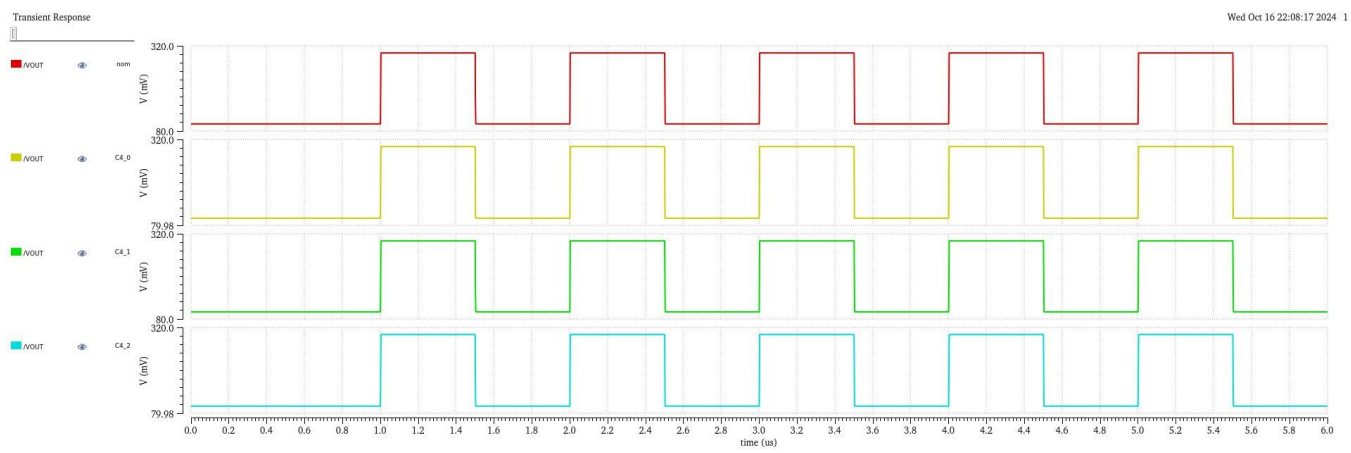
And why?



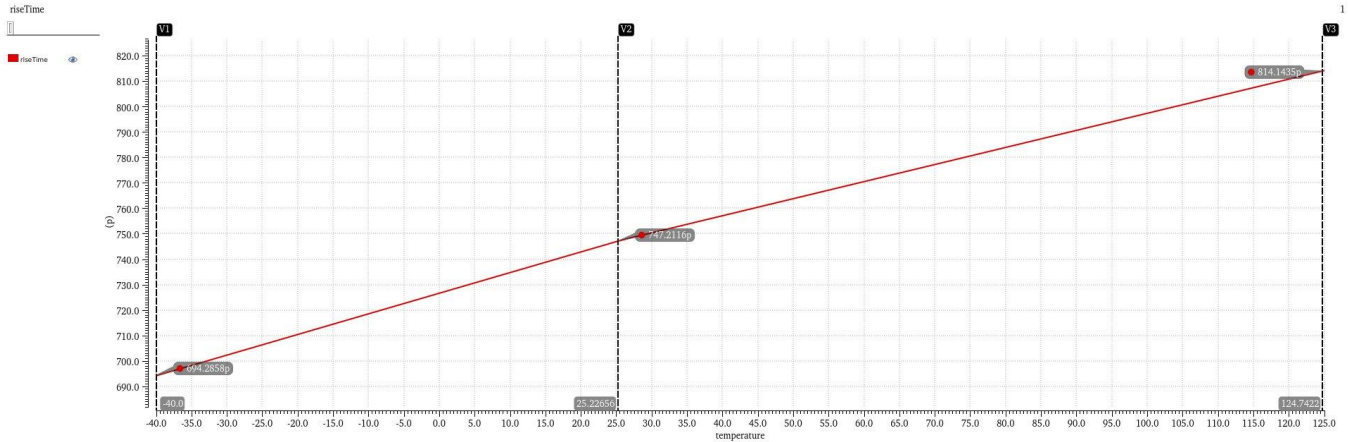


To understand why the rising time differs with corners I added these 3 extra graphs of the V_{th} (threshold voltage) , capacitance and ron of the NMOS with corners let's take corner 1 (ff_mim_ff) NMOS we will find that it has the lowest risetime risetime is associated with with switching speed of the NMOS so it's kind of trivial that ff corner with ff_mim produces the lowest rise time but we can also look at these variables to see that they are the lowest at this corner (low threshold voltage means transistor is on with lower value of V_{GS} and if we take a known value of rise time = 2.2 BW we'll find that lowering R and C will lower rise time the relation might not be exact be it's of course close) the second fastest rise time is corner (ss_mim_ff) indicating that the NMOS contribute more to the rise time change then nominal corner then the 3rd fastest is (ff_mim_ss) then the 4th fastest is (ss_mim_ss) corner the trend is mainly tied to the 3 variables that we discussed

• Plot V_{out} @ -40oC, 25oC and 125oC



• Plot Rising time of V_{out} @ -40oC, 25oC and 125oC and explain at each temperature



Increasing the temperature decreases the switching speed of the transistor you can observe this behavior in the summer while using your PC (V_{th} increases with temperature meaning the transistor needs higher gate voltage to switch on) and also the mobility is degraded with increase of temperature due to lattice vibrations .

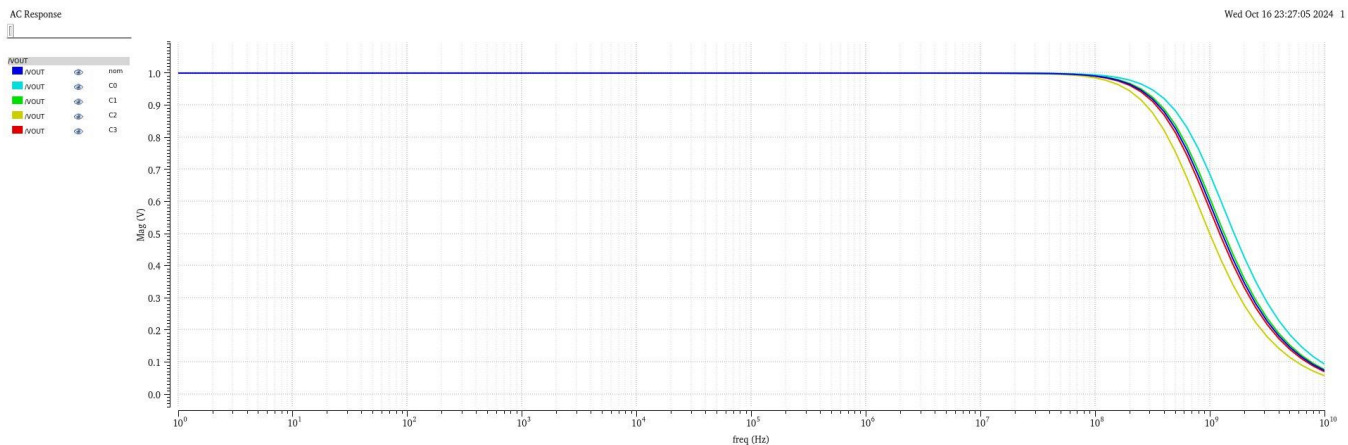
ii. AC Response

• $V_G = V_{dd}$

• $V_{in,dc} = 200mV$ & $V_{in,ac}=1V$

• Perform AC analysis from 10Hz to 10GHz with 10 points per decade

• Plot $V_{out,ac}$ across corners and explain at each corner what happens? And why?



The trend goes as follows in terms of largest bandwidth to lowest 1- (ff_mim ff corner clear first)

2- (ss_mim ff slightly higher than nominal) 3- nominal 4- (ff_mim ss slightly lower than nominal)

5- (ss_mim ss clear last) we could have expected this trend from the rise time graphs ; for the first corner (ff_mim ff) the threshold voltage is lowest , the cap is lowest and r_{on} of NMOS is lowest the bandwidth is inversely proportional to both cap value and resistance value as for the 2nd corner(ss_mim ff) variations

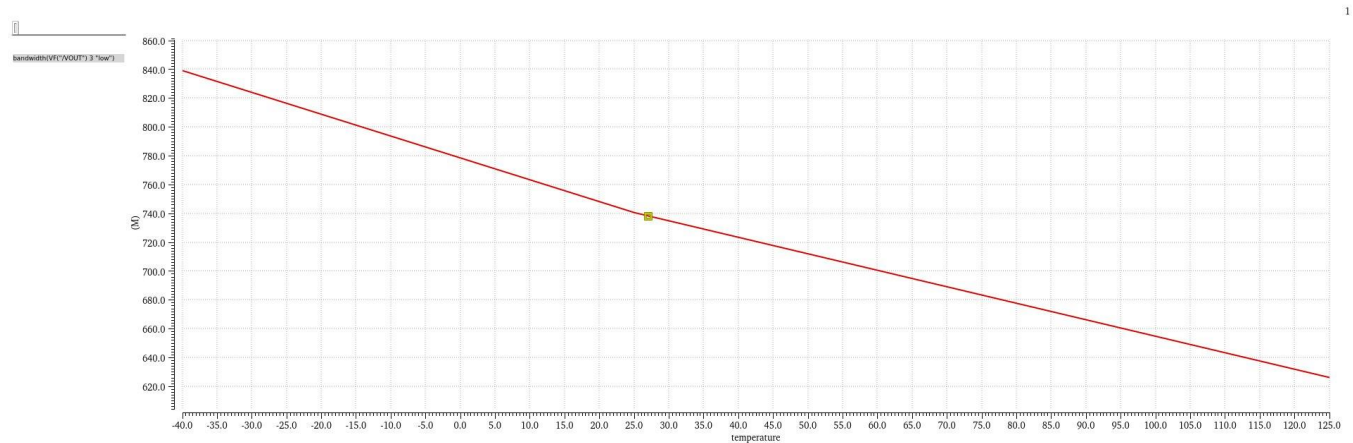
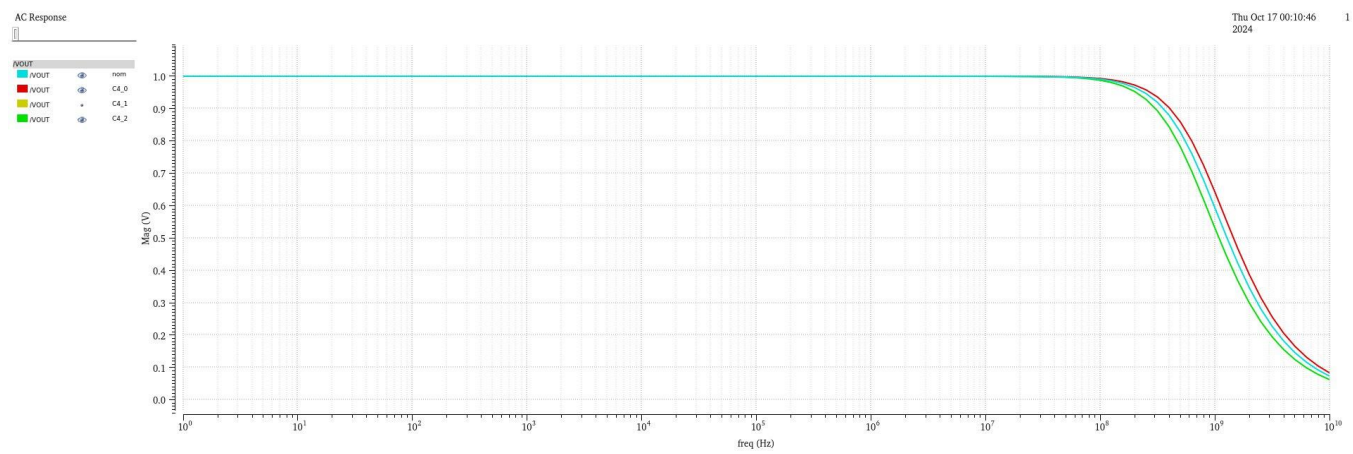
of the NMOS affecting the BW will overtake the lack that occurs in the cap value and overtake the BW of the nominal corner

For the third corner (ff_mim ss) despite the lowering of cap value the slowing down of the NMOS will end up making the BW of this corner slightly lower than nominal BW

NOTE: both these corners can pretty much be treated as nominal BW since the variation is not very big

For the fourth corner (ss_mim ss) both the cap value and ron and threshold voltage will increasing the rise time and lowering the bandwidth of the circuit

• Plot Vout,ac @ -40oC, 25oC and 125oC and explain at each temperature what happens? And why?

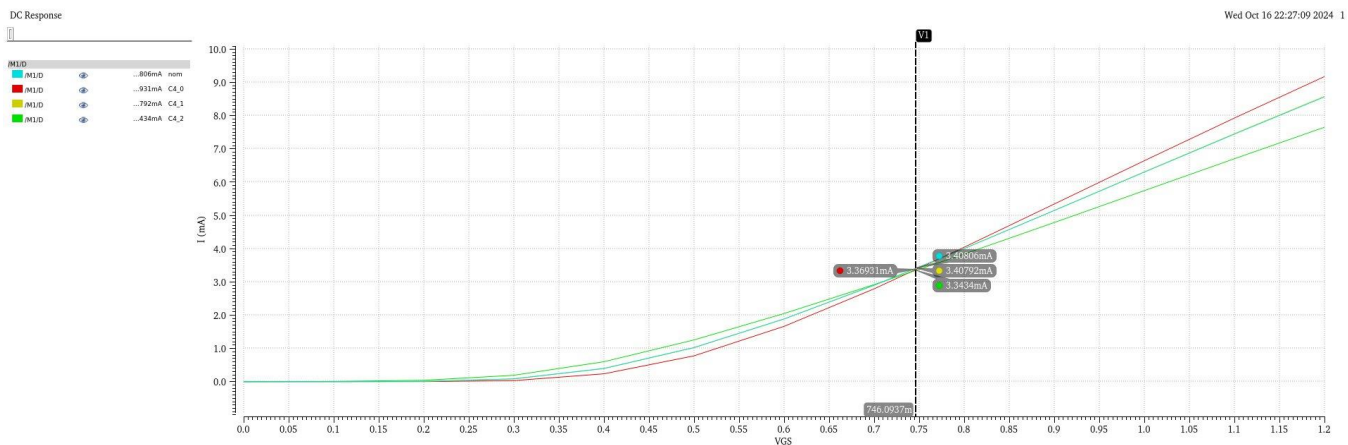


The trend is very clear here the bandwidth will simply decrease with the increase of temperature this can be attributed to the As the temperature rises, the decreased carrier mobility, increased

threshold voltage, larger parasitic capacitances, and higher resistances result in a lower cutoff frequency and thus reduced bandwidth

B) It is required to simulate MOSFET alone and plot ID vs VGS

- Plot ID vs VGS @ -40oC, 25oC and 125oC
- The three curves should intersect at certain point, what is the VGS at intersection?



Around $V_{GS} = 0.75 \text{ V}$

- Before intersection, At which temperature is the highest current and why?

Before intersection the lowest temperature MOSFET has the highest current this is mainly due to the fact the mobility of electrons will be higher due to increase of scattering of electrons in the lattice at higher temperatures.

- After intersection, At which temperature is the highest current and why?

after intersection another competing factor with the electron mobility takes over which is the V_{th} after the point of balance at around $V_{GS} = 0.75 \text{ V}$ the effect of lowering V_{th} with temperature takes over causing the current to be higher for higher temperatures

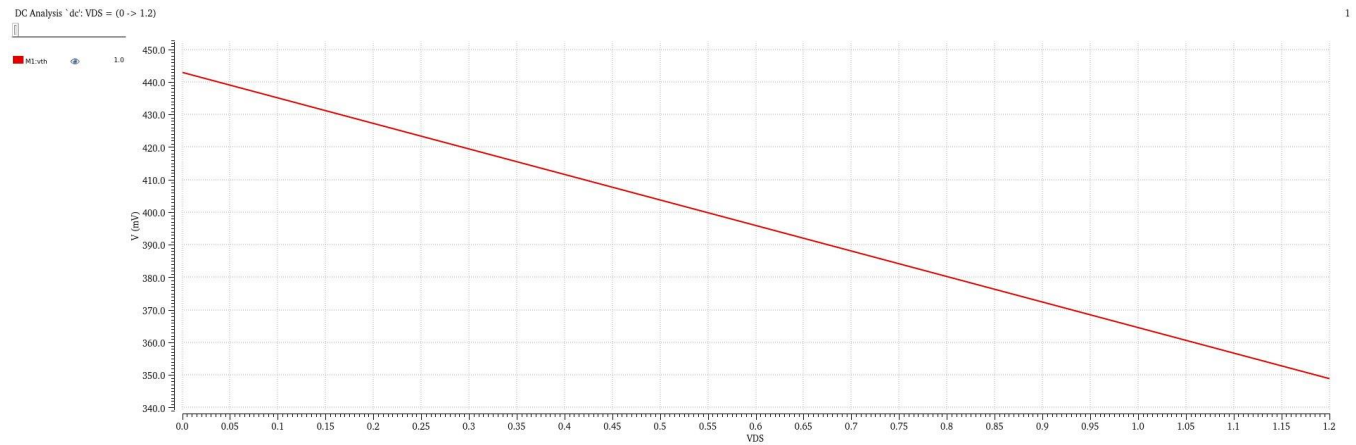
Part II: CMOS Short Channel Effects and temperature dependence

It is required to simulate MOSFET alone and plot V_{th} vs V_{DS}

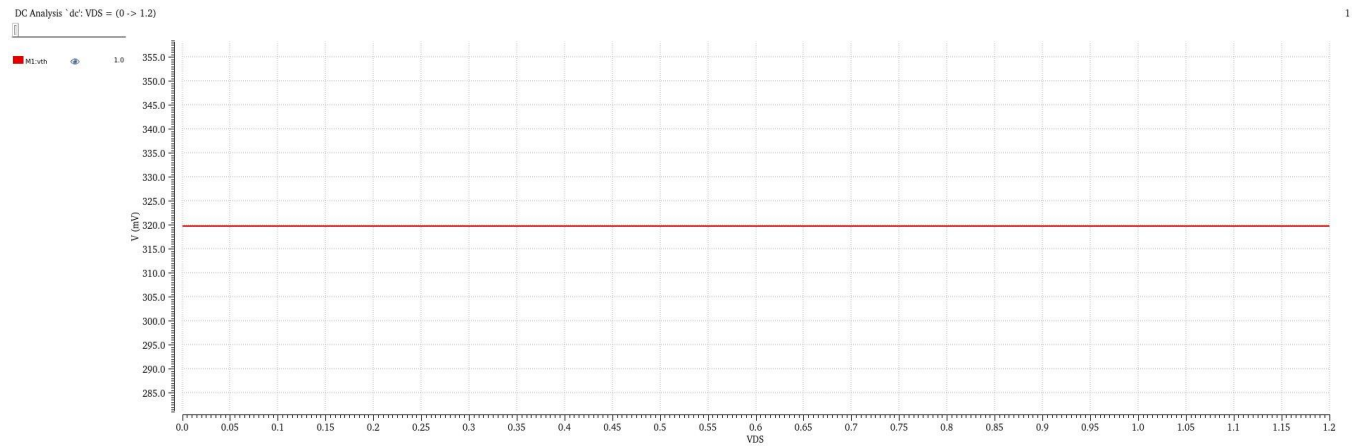
Plot V_{th} vs V_{DS} at the following channel Lengths:

• $L=60\text{nm}$ • $L=2\mu\text{m}$

$L=60\text{nm}$



$L=2\mu\text{m}$



Explain what happens at each channel length? and why? and identify the short channel effect.

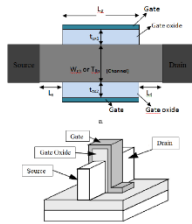
For the short channel MOSFET it is clear that the threshold voltage keeps decreasing while increasing V_{DS} this is mainly due to DIBL (drain induced barrier lowering) while for the long channel MOSFET the threshold voltage is pretty much constant across V_{DS} variation : the short channel effect DIBL is more apparent for short channel MOSFET since the lowering

The traditional CMOS technology was approaching its physical limits and many challenges arose from this like for example : the short channel effects (the gate control over current was getting weaker) many phenomena arose from that like DIBL , GIDL and threshold roll-off

Current leakage like gate current leakage and subthreshold leakage were increasing

Scaling down transistors increases the number of transistors per unit area, leading to higher power density. This results in heat dissipation challenges, causing thermal management issues.

What is FinFET (Fin Field-Effect Transistor): point FINFET in operation works just like a regular MOS transistor except that the ordinary structure has been folded along the gate length direction.



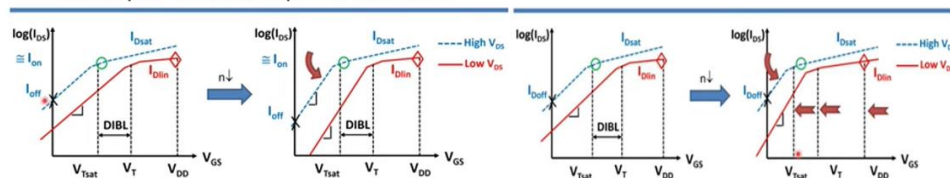
In logic we generally look to improve the speed of the device while also trying to decrease the power consumption so in general we can say that we are looking the decrease the V_{dd} ,increase the on current and decrease the off/leakage current of the device

What we find and what we will talk about more in the operation of the device is that there's always a tradeoff while changing the factors that affect these parameters and one key parameter is $n =$

$\frac{C_{ox} + C(parasitic)}{C_{ox}}$ a parameter that is affected by the capacitance indicates or points to why a change of the geometry is made

$$I_{off} = I_{Do} e^{\frac{qv_{eff}}{nkt}}, \text{ in a planar CMOS } n=1.5$$

FinFET performance: Impact of Reduced n



In the first picture we can look to keep the same threshold voltage that will result in a high slope and lower off current while the keeping the same on current

In the second picture we lower the threshold voltage that will result in keeping the off current the same as before but we can lower the V_{DD} since we have lowered the V_t and V_{DD} plays a major role in reducing the power consumption

The 3D structure of the FinFET enhances gate control over the channel. This reduces short-channel effects like DIBL and threshold voltage variations, allowing for better scalability to smaller nodes while maintaining performance.

Reduced Leakage Currents:

FinFETs provide stronger gate control, which helps suppress leakage currents, particularly subthreshold and gate leakage. The multi-gate architecture ensures that the channel is better confined, reducing leakage pathways.

Lower Power Consumption:

Due to the improved control over the channel, FinFETs can operate at lower voltages without sacrificing performance. This helps in reducing dynamic and static power consumption, making FinFETs ideal for low-power applications.

FinFETs offer higher drive current and faster switching speeds due to their lower effective channel length and reduced parasitic effects. This leads to improved performance for high-speed applications.